Homework:

Malware Classification based on *Federated Learning* *under Differentially-Private Protection*

Malware Classification based on *Federated Learning under Differentially-Private Protection*

- Replace the machine learning method in Homework 5 with Federated Learning *combined with Differentially-Private Protection*.

  - Some References:
    - P. Kairouz, Ziyu Liu, and Thomas Steinke. "The distributed discrete gaussian mechanism for federated learning with secure aggregation." *International Conference on Machine Learning*. PMLR, 2021. => Suggested (because it's newer result)
    - K. Wei, et al. "Federated learning with differential privacy: Algorithms and performance analysis." *IEEE Transactions on Information Forensics and Security* 15 (2020): 3454-3469.
    - … (Too many papers for this hot topic)

  - Some *Basics*:
    - M. Abadi, et al. "Deep learning with differential privacy." *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*. 2016.

      => Only for Deep Learning, *not for Federated Learning*. But there are some important basic ideas in this paper.

Hints

1. Repeat Homework 5, but replace the machine learning method with Federated Learning *combined with Differentially-Private Protection*.
    - You could apply this implementation code for malware analysis:
        - https://github.com/google-research/federated/tree/master/distributed_dp
            - Other resource: https://www.tensorflow.org/federated/api_docs/python/tff/learning/ddp_secure_aggregator

2. Training and testing by machine learning model for classify malware (image).

Upload the result to Moodle

- Upload your code and the ***detailed*** reports to Moodle.
  - Code: You must add some comments in your code for easy understanding.
  - Report Files: Document (Word)
  - Zipped to a file. The file name should be "*Team Name*_HW6.zip"

Appendix:
Other Reference for **Deep Learning** with Differential Privacy (not for ***Federated Learning***)

- GitHub - lingyunhao/Deep-Learning-with-Differential-Privacy: An implementation of Deep Learning with Differential Privacy
- Google's *TensorFlow Privacy:* GitHub - tensorflow/privacy: Library for training machine learning models with privacy for training data
    - Introducing TensorFlow Privacy: Learning with Differential Privacy for Training Data — The TensorFlow Blog

- Though the above code are only for Deep Learning, and *not for Federated Learning (distributed environment)*.

However, you could check this, and even possibly expand this to federated learning manually, if you cannot use the GitHub code mentioned in previous page.